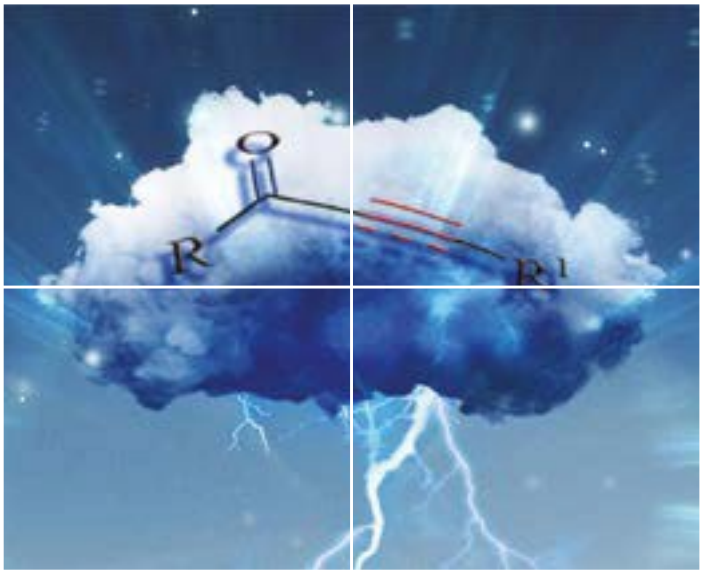
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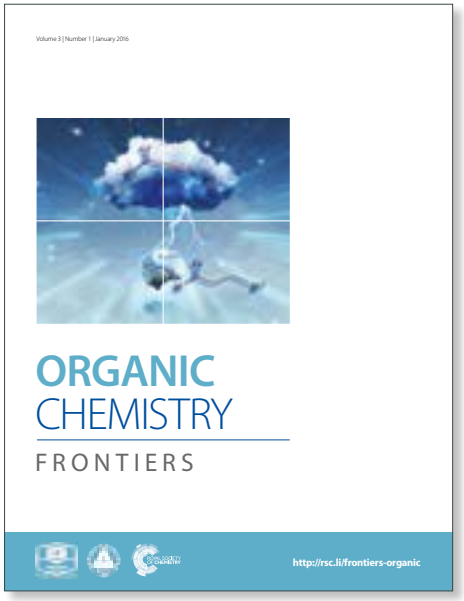
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E-Selective *N*-Heterocyclic Carbene-Catalyzed Reaction of Aldehydes and Butadienoates: Effect of Water and Chloroform as the Proton Shuttle

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As we know, there are many reports on the reactions of the Breslow intermediate with electron-deficient alkenes or alkynes. However, the reactions with allenes are much less studied. In this study we observed that *N*-heterocyclic carbene-catalyzed reaction of aldehydes and butadienoates afforded (*E*)-4-oxo-2-butenoates highly stereoselectively. It was observed that addition of water is critical for the observed high stereoselectivity. Mechanistic studies showed that the solvent chloroform acted as the proton shuttle involved in the formation of the Breslow intermediate, protonation, and deprotonation for this carbene-catalyzed reaction. Some key intermediates such as the Breslow intermediate and its mono hydrate have been detected and characterized by MS studies, supporting the proposed mechanism.

Introduction

The Breslow intermediate formed from the reaction of simple aldehydes and carbenes, firstly disclosed in 1958,[1] was ubiquitously involved in many *N*-heterocyclic carbene-catalyzed reactions. Ever since its discovery, studies on the mechanism have always been on a par with development of corresponding new transformations.[2] A proton transfer process, mediated by base,[3] H_2O,[3] conjugate acid,[4] or intramolecular 1,2-hydrogen shift,[5] and even intermolecular proton transfer involving two NHC/aldehyde adducts,[4a,6] after the primary 1,2-addition of *N*-heterocyclic carbene with aldehyde in the formation of the Breslow intermediate have been reported. Apart from that, the spiroepoxide intermediate[7] and radical mechanism[8] were also proposed. On the other hand, in 2012, Berkessel et al. pioneeringly characterized the 2,2-diamino enols by NMR spectroscopy and it was proved that the α-proton of aldehyde is proven to be the source of hydroxylic proton in the Breslow intermediate (Scheme 1A, eq. 1).[9,10] Recently, we reported a *N*-heterocyclic carbene-catalyzed cyclization reaction of aldehydes and 1,2-allenones affording cyclopent-2-enone-4-ols **S1** with high diastereoselectivity referring to the two chiral centers.[11] The intermediate **Int-I** with *Z*-configuration formed after the addition of the Breslow intermediate with 1,2-allenone was believed to be crucial for the subsequent cyclization (Scheme 1B, eq. 2). What's more, when α-deuteriobenzaldehyde was submitted to the reaction, deuterated product was afforded smoothly, confirming the proton transfer process as in reported literatures.[9] Interestingly, when butadienoates were used as the coupling partners with the Breslow intermediate in the presence of water, (*E*)-4-oxo-2-butenoates (*E*)-**3** were formed highly stereoselectively: the formation of stereoisomer (*Z*)-4-oxo-2-butenoate (*Z*)-**3** and double bonds isomer **4** are negligible. These results illustrated the entirely different intermediate **Int-II** with the *E*-configuration after the addition of the Breslow intermediate with butadienoates (Scheme 1B, eq. 3). In addition, besides the Breslow intermediate formed in the similar manner as that Berkessel et al. reported (Scheme 1C, path a),[9] we also observed a rather unexpected phenomenon: the primary adduct of *N*-heterocyclic carbene with aldehyde was most probably protonated with the solvent-chloroform, led to the transformation with chloroform as the proton shuttle (Scheme 1C, eq. 4) and the formation of the monohydrated Breslow intermediate is responsible for the observed stereoselectivity as supported by MS studies (Scheme 1C, eq. 5). We wish to report here these observations.

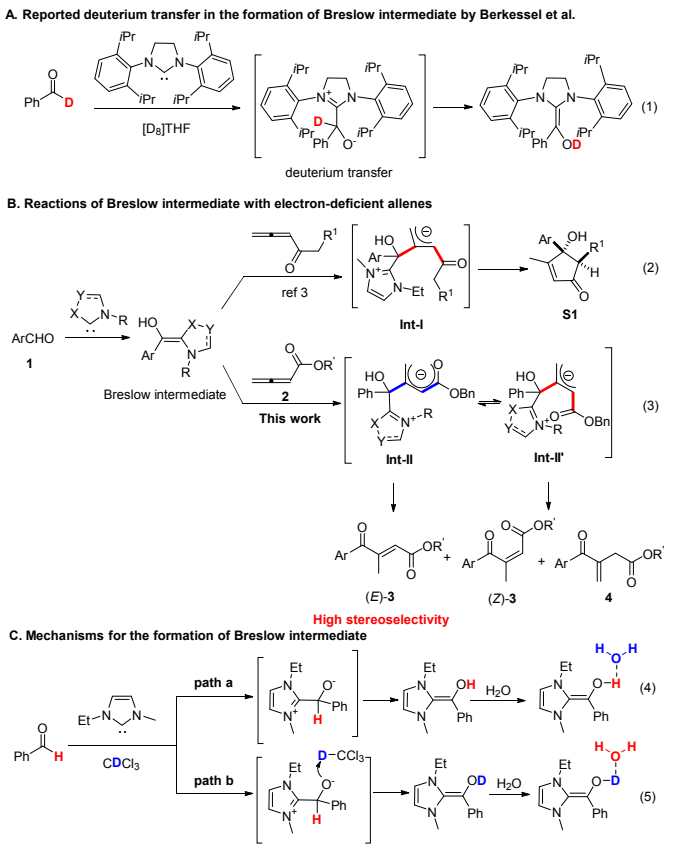
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Electronic Supplementary Information (ESI) available: Experimental section, characterization of all the compounds, and copies of ^{1}H NMR, ^{13}C NMR spectra of selected compounds. See DOI: 10.1039/x0xx00000x

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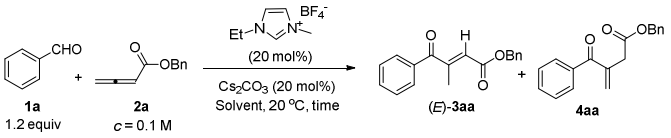
Scheme 1. Formation of the Breslow intermediate and our observation.

Results and discussion

Initially, we attempted the reaction of benzaldehyde **1a** with butadienoate **2a** in the presence of 20 mol% 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIMBF$_4$) and 20 mol% Cs_2CO_3 in toluene at -10 °C. Disappointingly, no product was obtained with 84% recovery of **2a** (Table 1, entry 1). Gladly, the formation of **3aa** was observed in dioxane or THF at 20 °C with full consumption of **2a**, however, the stereoselectivity is poor (Table 1, entries 2 and 3). When DCM was used as the solvent, (*E*)-**3aa** was formed as the sole product in 42% yield with 22% recovery of **2a** (Table 1, entry 4). At last, we were very pleased to find that running the reaction in commercial $CHCl_3$ under 30 mol% of Cs_2CO_3 gave the best results: the yield of (*E*)-**3aa** was 61% with exclusive *E* selectivity while **4aa** was formed in only 1% yield with full conversion of **2a** (Table 1, entry 5).

When 1.8 equiv of **1a** were used, the stereoselectivity of **3aa** was somewhat improved. Thus, the reaction was optimized with 1.8 equiv of **1a**. Quite unexpectedly, control experiments showed that reactions carried out in commercial $CHCl_3$ and anhydrous $CHCl_3$ (distilled from CaH_2 under N_2 before use) gave rather different results: the stereoselectivity and yield of **3aa** in anhydrous $CHCl_3$ dropped to 92/8 and 61%, and **4aa** was formed in 7% yield, which indicated the potential role of water in the commercial $CHCl_3$ (Table 2, entries 1 and 2).

Table 1. Optimization of reaction conditions for the highly stereoselective synthesis of (*E*)-**3aa**[a]

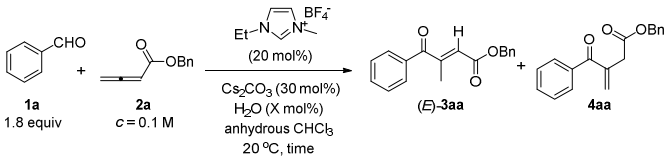


Entry	Solvent	time (h)	Yield[b] (%)			Recovery of **2a**[b] (%)
			3aa	*E*/*Z*	**4aa**	
1[c]	toluene	22	-	-	-	84
2	THF	11.5	59	73/27	10	-
3[d]	dioxane	17	63	89/11	8	-
4	DCM	21.5	42	>99/1	1	22
5[e]	$CHCl_3$[f]	18.3	61	>99/1	1	-

[a] Reaction conditions: A solution of NHC precursor and Cs_2CO_3 in solvent (1 mL) was stirred at 20 °C for 10 min, then **1a**, solvent (0.5 mL), 0.2 mmol of **2a**, and solvent (0.5 mL) were added sequentially. The resulting mixture was then stirred at T °C for the time indicated in the Table. [b] Determined by NMR analysis of the crude product using dibromomethane (for entries 1-3) and 1,3,5-trimethylbenzene (for entries 4-5) as the internal standard. [c] The reaction was conducted at -10 °C. [d] The reaction was conducted using 0.4 mmol of **2a**. [e] The reaction was conducted using 1.5 equiv **1a** and 30 mol% of Cs_2CO_3. [f] $CHCl_3$ was purchased from Sinopharm Chemical Reagent Co., Ltd and used as received.

Then, different amounts of water were deliberately added to the anhydrous $CHCl_3$ (Table 2, entries 3-7). To our delight, an improved yield of **3aa** and stereoselectivity were observed when 25 mol% of H_2O was added (Table 2, entry 3). Finally, adding 40 mol% H_2O was proven to be the best: the yield of **3aa** was 66%, the ratio of (*E*)-**3aa**/(*Z*)-**3aa** was >99/1, and **4aa** was produced in only 2% yield (Table 2, entry 5).

Table 2. Further optimization of the reaction conditions-effect of water[a]



Entry	X	time (h)	Yield[b] (%)			Recovery of **2a**[b] (%)
			3aa	*E*/*Z*	**4aa**	
1[c,d]	0	17.3	65	>99/1	2	2
2[d]	0	20	61	92/8	7	-
3	25	17.5	67	99/1	4	-
4	35	17	66	>99/1	3	-
5	40	17.5	66	>99/1	2	<0.5
6	45	17.5	58	>99/1	1	5
7	50	17.3	46	>99/1	<0.5	25

[a] Reaction conditions: A solution of NHC precursor and Cs_2CO_3 in $CHCl_3$ (2 mL) was stirred at 20 °C for 10 min, then **1a**, $CHCl_3$ (1 mL), 0.4 mmol of **2a**, $CHCl_3$ (1 mL), and H_2O were added sequentially. The resulting mixture was then stirred at 20 °C for the time indicated in the Table. [b] Determined by NMR analysis of the crude product using 1,3,5-trimethylbenzene as the internal standard. [c] $CHCl_3$ was purchased from Sinopharm Chemical Reagent Co., Ltd and used as received. [d] The reaction was conducted using 1 mmol of **2a**.

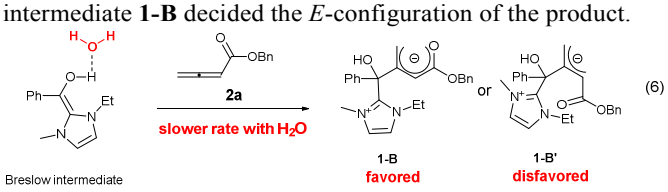
We inferred that the water added most probably formed a hydrogen bond with the OH unit in the Breslow intermediate generated *in situ*, which would greatly reduce its reactivity

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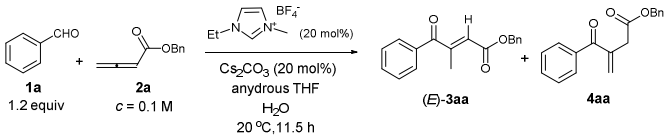
towards butadienoate **2a**, finally yielding the thermodynamically more stable intermediate **1-B** highly stereoselectively (eq. 6). Thus, the highly selective formation of intermediate **1-B** decided the *E*-configuration of the product.



(6)

To verify our hypothesis, water was also added to the reactions carried out in anhydrous THF (distilled from Na wire under N_2 before use). Again, when 40 mol% of water was added, the yield of **3aa** was increased to 64%, together with an improved *E*/*Z* stereoselectivity to 89/11 (Table 3, entry 2). Furthermore, the *E*/*Z* stereoselectivity could be improved to as high as 98/2 when 1 equiv of water was added, together with 12% recovery of **2a** (Table 3, entry 3).

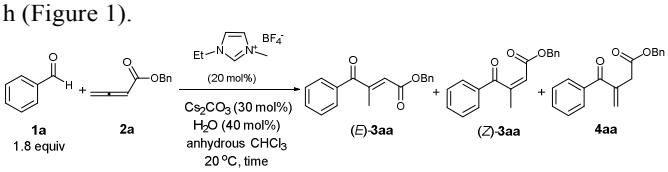
Table 3. Reactions carried out in anhydrous THF with water as additive[a]



Entry	H_2O (mol %)	Yield[b] (%)			Recovery of **2a**[b] (%)
		3aa	*E*/*Z*	**4aa**	
1	0	59	73/27	10	-
2	40	64	89/11	10	0.5
3	100	59	98/2	5	12

[a] A solution of NHC precursor and Cs_2CO_3 in THF (1 mL) was stirred at 20 °C for 10 min, then **1a**, THF (0.5 mL), 0.2 mmol of **2a**, THF (0.5 mL), and H_2O were added sequentially. The resulting mixture was then stirred at 20 °C for 11.5 h. [b] Determined by NMR analysis of the crude product using dibromomethane (for entry 1) and 1,3,5-trimethylbenzene (for entries 2-3) as the internal standard.

With the aim of examining the formation sequence of (*E*)-**3aa**, (*Z*)-**3aa**, and **4aa**, and observing whether inter-conversion exists among these three compounds, the reaction was monitored by NMR at different reaction time. As we can see, (*E*)-**3aa** was formed directly in the first 7 h without the formation of (*Z*)-**3aa** and **4aa**. The formation of (*E*)-**3aa** turned to be very slow with the remaining 7% of **2a**. Meanwhile, (*Z*)-**3aa** and **4aa** were observed along with a prolonged reaction time. At last, (*E*)-**3aa** was formed in 63% yield, (*Z*)-**3aa** was formed in <1% yield, and **4aa** was formed in 2% yield after 19 h (Figure 1).



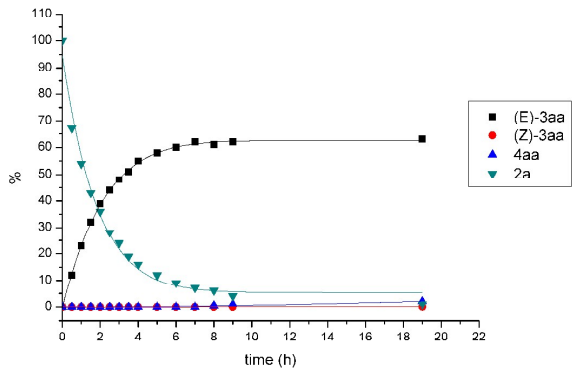
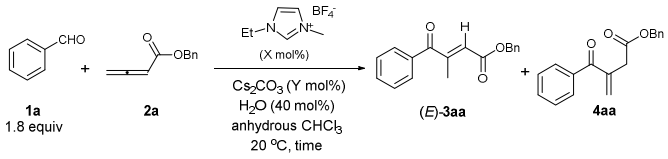


Figure 1. Monitoring of the reaction.

In order to unveil the mechanism, we did some control experiments: **3aa** couldn't be formed in the absence of Cs_2CO_3 (Table 4, entry 1) and/or $EMIMBF_4$ (Table 4, entries 2 and 3). Thus, *N*-heterocyclic carbene is obviously crucial in this transformation.

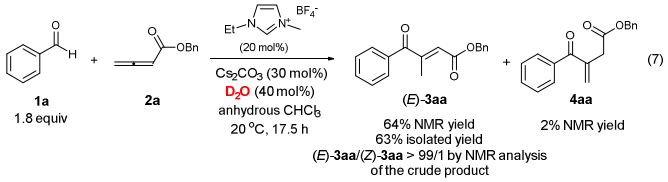
Table 4. Control experiments[a]



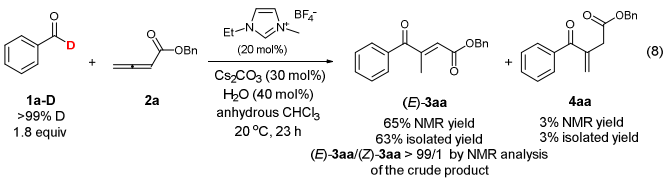
Entry	X/Y	time (h)	Yield[b] (%)			Recovery[b] (%)	
			3aa	*E*/*Z*	**4aa**	**1a**	**2a**
1	20/0	22	-	-	-	70	96
2	0/30	21.5	-	-	-	75	95
3	0/0	21.8	-	-	-	57	95

[a] Reaction conditions: A solution of NHC precursor and/or Cs_2CO_3 in $CHCl_3$ (2 mL) was stirred at 20 °C for 10 min, then **1a** (1.8 equiv), $CHCl_3$ (1 mL), 0.4 mmol of **2a**, $CHCl_3$ (1 mL), and H_2O were added sequentially. The resulting mixture was then stirred at 20 °C for the time indicated in the Table. [b] Determined by NMR analysis of the crude product using 1,3,5-trimethylbenzene as the internal standard.

Then, H_2O was replaced with D_2O, interestingly, non-deuterated (*E*)-**3aa** was formed (eq. 7). Quite surprisingly, even the reaction of deuterated benzaldehyde **1a-D** with **2a** also yielded non-deuterated (*E*)-**3aa** (eq. 8).



(7)



(8)

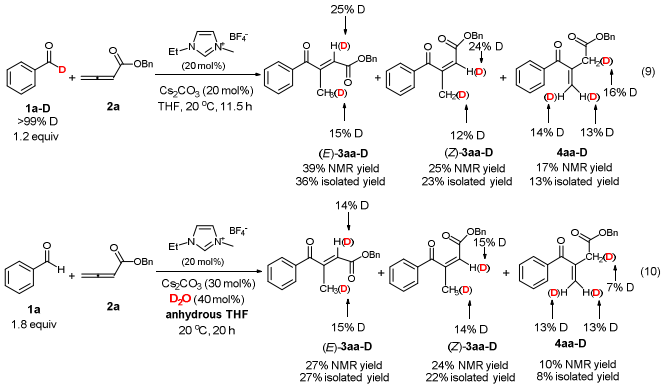
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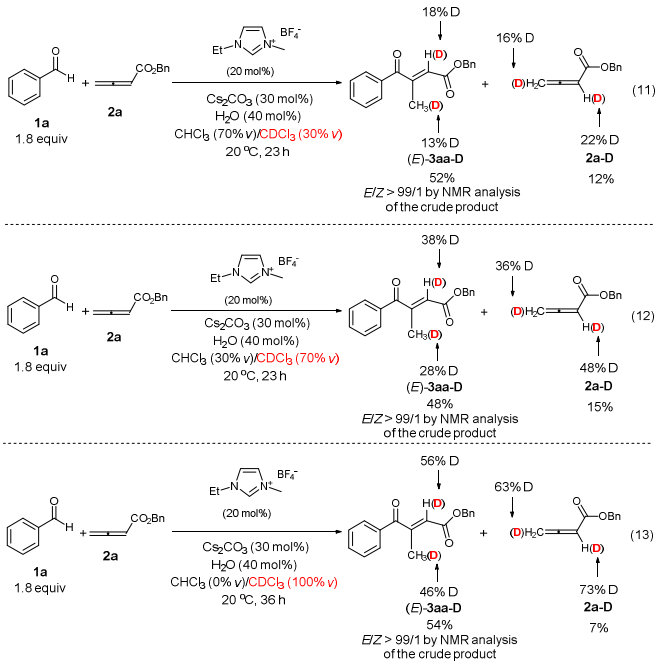
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A parallel experiment was conducted using **1a-D** in anhydrous THF. However, three different deuterated products were obtained smoothly: deuterated (*E*)-**3aa-D** was formed in 36% yield with 25% D-incorporation at the 2-position olefinic proton and 15% D-incorporation at the 3-position methyl group; deuterated (*Z*)-**3aa-D** was formed in 23% yield with 24% D-incorporation at the 2-position olefinic proton and 12% D-incorporation at the 3-position methyl group; deuterated **4aa-D** was formed in 13% yield with 16% D-incorporation at the 2-position methylene, 14% and 13% D-incorporation at the olefinic proton of 4-position methylene group respectively (eq. 9). Another parallel reaction of **1a** and **2a** was conducted in the presence of D_2O (40 mol%) in anhydrous THF. Again, three deuterated products were produced: deuterated (*E*)-**3aa-D** in 27% yield with 14% D-incorporation at the 2-position olefinic proton and 15% D-incorporation at the 3-position methyl group, deuterated (*Z*)-**3aa-D** in 22% yield with 15% D-incorporation at the 2-position olefinic proton and 14% D-incorporation at the 3-position methyl group, and deuterated **4aa-D** in 8% yield with 7% D-incorporation at the 2-position methylene, 13% each of D-incorporation at the two terminal olefinic protons of 4-position methylene group respectively (eq. 10). Obviously, the reaction mechanism in $CHCl_3$ was quite different from that in THF.

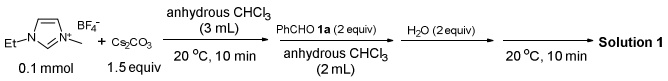


The reaction of **1a** with **2a** was then carried out in mixed solvents of $CHCl_3$/$CDCl_3$ with different volume ratios under the standard conditions. As a result, deuterated (*E*)-**3aa-D** was indeed formed in moderate yields, together with some recovery of deuterated butadienoate **2a-D**. It was clear to see that the D-incorporation increased in both the product (*E*)-**3aa-D** and the recovered allenoates **2a-D** with more $CDCl_3$ (Scheme 2).



Scheme 2. Control experiments in mixed solvents of $CHCl_3$/$CDCl_3$ with different volume ratios.

SAESI-MS was further utilized to capture any Breslow intermediate formed *in situ*.[12] 0.1 mmol of $EMIMBF_4$ and 0.15 mmol of Cs_2CO_3 were stirred in anhydrous $CHCl_3$ (3 mL) at 20 °C for 10 min under Ar atmosphere. Then 0.2 mmol of **1a**, anhydrous $CHCl_3$ (2 mL), and 0.2 mmol of H_2O were added sequentially under Ar atmosphere. The resulting mixture was stirred at 20 °C. After 10 min, the reaction mixture was submitted to SAESI-MS analysis. This reaction **solution 1** was injected by a 500-μL air-tight syringe at a speed of 5 μL/min to SAESI-MS. The assisting solvent of MeOH was injected by another 500-μL air-tight syringe at a speed of 5 μL/min to SAESI-MS. The signal of protonated Breslow intermediate **IM1** (or its keto tautomer **IM1'**) was detected at *m/z* 217. Most importantly, signal for monohydrated Breslow intermediate, namely ion **IM2** was also observed (Figure 2), which confirmed the mechanism we proposed in eqn (6). The signal of m/z 235 could only be detected when MeOH was used as assisting solvent but not with CH_3CN as the only assisted solvent (For details, see Figure S1 and S2 in the Supporting Information). We considered that it was helpful to form the hydrogen bond when MeOH was used as assisting solvent. Furthermore, this peak of m/z 235 did not exist in the background peaks of solvent $CHCl_3$, MeOH, and CH_3CN. Thus, this intermediate was formed in the reaction.



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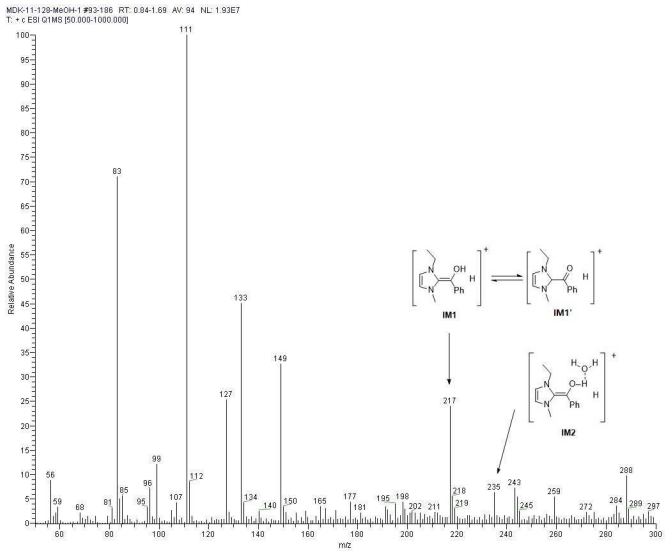


Figure 2. SAESI-MS spectrum of the reaction **solution 1** using MeOH as the assisted solvent.

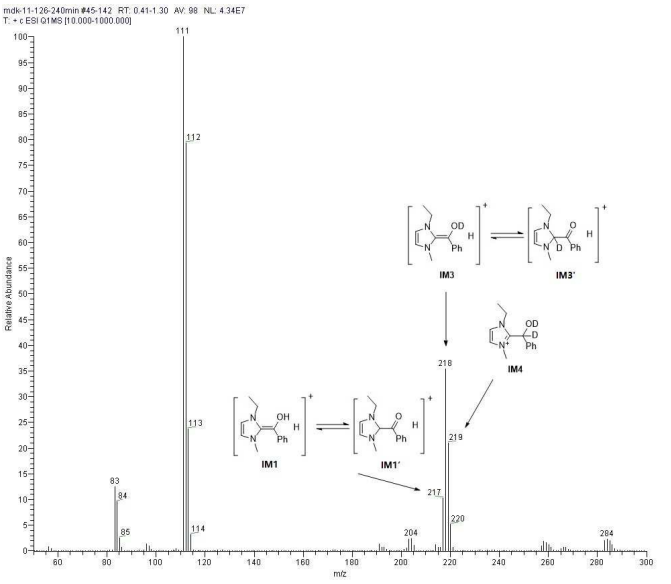
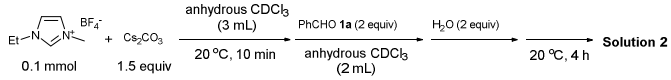


Figure 3. SAESI-MS spectrum of the reaction **solution 2** using anhydrous acetonitrile as the assisted solvent.

Then $EMIMBF_4$ (0.1 mmol) and 0.15 mmol of Cs_2CO_3 were stirred in anhydrous $CDCl_3$ (3 mL) at 20 °C for 10 min under Ar atomosphere. Then 0.2 mmol of **1a**, anhydrous $CDCl_3$ (2 mL), and 0.2 mmol of H_2O were added sequentially under Ar atmosphere. The resulting mixture was stirred at 20 °C. After 4 h, the reaction mixture was submitted to SAESI-MS analysis. The reaction **solution 2** was injected by a 500-μL air-tight syringe at a speed of 5 μL/min to SAESI-MS. The assisting solvent, anhydrous acetonitrile, was injected by another 500-μL air-tight syringe at a speed of 5 μL/min to SAESI-MS. Gladly, the signal of protonated D-Breslow intermediate **IM3** (or its keto tautomer **IM3'**) at *m/z* 218 was detected, confirming the formation of deuterated Breslow intermediate in $CDCl_3$. In fact, we observed that the signal intensities of the target ions *m/z* 218 and *m/z* 219 turned to be stronger as the prolonging reaction time (For details, see Figures S3-S8 in the Supporting Information). At the same time, the peak of *m/z* 217 indicated the reaction pathway for the formation of the Breslow intermediate as Berkessel et al. reported[9] (Scheme 1C, path a). At last, we inferred that the peak of *m/z* 219 should be ion **IM4**, formed after deprontonation at the benzylic position and reprotonation with $CDCl_3$ (Figure 3).



Based on these mechanistic studies, we proposed a possible reaction mechanism for the formation of (*E*)-**3aa-D** in $CDCl_3$. *N*-Heterocyclic carbene generated *in situ* would react with benzaldehyde **1a** in the presence of water to form monohydrated Breslow intermediate **1-A**. Then intermeidate **1-A** would react with **2a** to form a delocalized anionic intermediate **1-B**[11,13] due to the steric effect. Intermediate **1-B** would be protonated with $CDCl_3$ to form intermediate **1-C** or **1-D**. Subsequently, in the presence of Cs_2CO_3 or trichloromethide carbanion, intermediate **1-C** would be deprotonated at the hydroxyl group to give intermediate **1-F**. Subsequent elimination of *N*-heterocyclic carbene would generate (*E*)-**3aa-D** and regenerate the carbene (Scheme 3, path a). The active methylene unit in intermediate **1-D** would be deprotonated to yield intermediate **1-E** by *N*-heterocyclic carbene[14] or Cs_2CO_3. Then intermediate **1-E** was protonated by $CDCl_3$ to from intermediate **1-C'**. In the presence of Cs_2CO_3 or trichloromethide carbanion, intermediate **1-C'** would also be transferred to intermediate **1-F** (Scheme 3, path b).

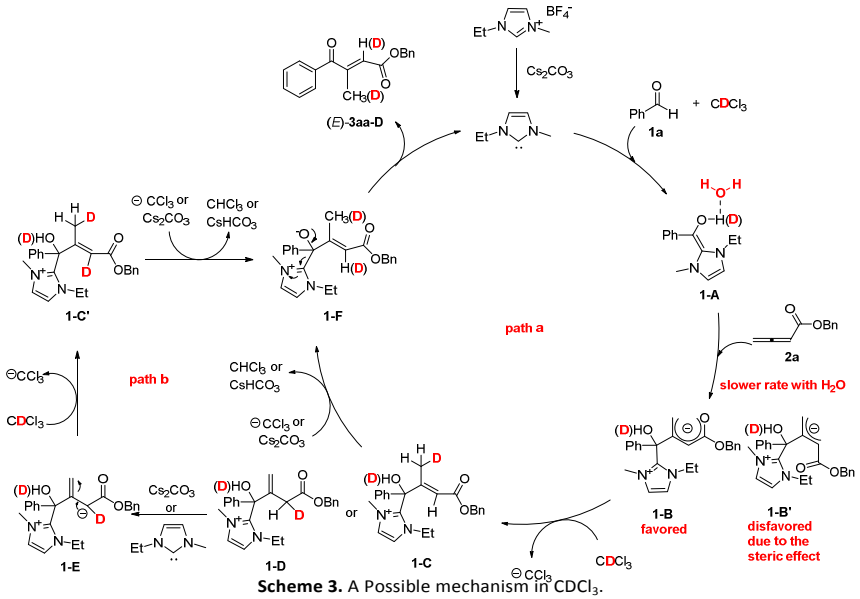
(*E*)-4-Oxo-2-butenoates have been proven to be important motifs of some bioactive compounds, natural products, and macrocyclic antibiotics, such as pyrenophorin and vermiculin.[15] They are also widely used as useful synthon in many organic transformations, such as Diels-Alder reaction,[16] cyclization,[17] cyclopropanation,[18] addition reactions as Michael acceptors,[19] etc., affording a series of carbocycles, heterocycles and other polyfunctional structures. Thus, these compounds are attracting continuous interest. So far, many methods have been developed for the preparation of (*E*)-4-oxo-2-butenoates.[20] However, these methods usually suffer from unsatisfactory stereoselectivity, low atom economy, non-readily-available starting materials, using stoichiometric or toxic reagents in some cases. Therefore, the highly stereoselective synthesis of (*E*)-4-oxo-2-butenoates

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from readily available starting materials under milder reaction conditions with atom-economy, is still of current interest.



Scheme 3. A Possible mechanism in $CDCl_3$.

With the optimized conditions in hand, different aldehydes were tested. Gladly, Ar group could be phenyl (Table 5, entry 1), electron-withdrawing group substituted such as *p*-Br (Table 5, entry 2), *m*-Br (Table 5, entry 3), *p*-Ph (Table 5, entry 5), *p*-CN (Table 5, entry 6), *p*-COOMe (Table 5, entry 7), and *p*-$COCH_3$ (Table 5, entry 8), or electron-donating *p*-OMe substituted phenyl (Table 5, entry 9). What's more, 2-naphthaldehyde was also compatible (Table 5, entry 10). The reaction of heteroaromatic aldehydes, such as 2-furaldehyde and 2-thenaldehyde both proceeded smoothly (Table 5, entries 11-12). The (*E*)-**3**/(*Z*)-**3** selectivity was great than 97/3. Probably due to the steric effect, the reaction of 2-bromobenzaldehyde **1d** with **2a** gave (*E*)-**3da** in only 9% yield, together with 18% recovery of **2a** (Table 5, entry 4). The structure of the products was further confirmed by the X-ray single crystal diffraction study of (*E*)-**3ha**[21] (Figure 4).

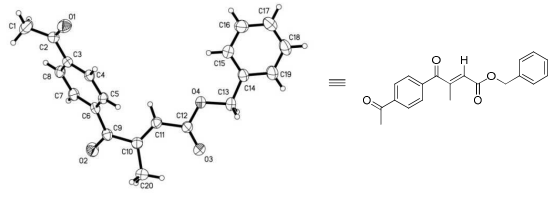
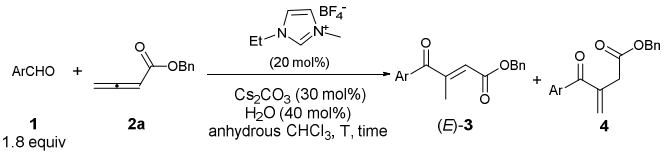


Figure 4. ORTEP representation of (*E*)-**3ha**.

Table 5. Substrate scope of aromatic aldehydes[a]



Entry	**1** Ar	T (°C)/ time (h)	Yield (%) (*E*)-**3**[b] (*E*/*Z*)[c]	**4**[c]
1	Ph (**1a**)	20/17	65 (**3aa**) (>99/1)	3
2	4-BrC_6H_4 (**1b**)	5/18.5	62 (**3ba**) (97/3)	2
3	3-BrC_6H_4 (**1c**)	5/18.7	66 (**3ca**) (97/3)	4
4[d]	2-BrC_6H_4 (**1d**)	20/40.7	9 (**3da**) (>99/1)	-
5	4-PhC_6H_4 (**1e**)	10/17	68 (**3ea**) (>99/1)	1
6[e]	4-NCC_6H_4 (**1f**)	20/43	30 (**3fa**) (97/3)	-
7	4-$MeO_2CC_6H_4$ (**1g**)	20/17.5	64 (**3ga**) (98/2)	1
8[f]	4-AcC_6H_4 (**1h**)	20/17.5	49 (**3ha**) (98/2)	-
9[g]	4-$MeOC_6H_4$ (**1i**)	50/36	35 (**3ia**) (>99/1)	-
10[h]	2-naphthyl (**1j**)	5/18.5	55 (**3ja**) (>99/1)	1
11	2-furanyl (**1k**)	20/16.5	61 (**3ka**) (97/3)	3
12[i]	2-thienyl (**1l**)	10/25	64 (**3la**) (>99/1)	1

[a] Reaction conditions: A solution of NHC precursor (20 mol%) and Cs_2CO_3 (30 mol%) in $CHCl_3$ (5 mL) was stirred at 20 °C for 10 min, then **1** (1.8 equiv), $CHCl_3$ (2.5 mL), 1 mmol of **2a**, $CHCl_3$ (2.5 mL), and H_2O (7.2 μL) were added sequentially. The resulting mixture was then stirred at T °C for the time indicated in the Table. [b] Isolated yield. [c] Determined by NMR analysis of the crude product using 1,3,5-trimethylbenzene as the internal standard. [d] 18% of **2a** was recovered. [e] 33% of **2a** was recovered. [f] 3% of **2a** was recovered. [g] 18% of **2a** was recovered. [h] 2% of **2a** was recovered. [i] 3% of **2a** was recovered.

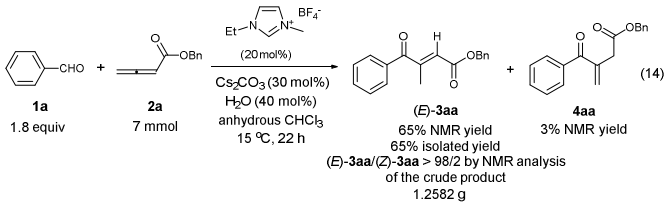
The reaction of 7 mmol of **2a** with **1a** at 15 °C afforded 1.2582 g of (*E*)-**3aa** with >98/2 *E*/*Z* stereoselectivity (eq. 14).

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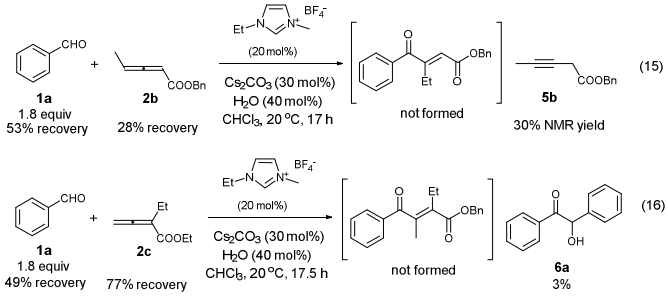
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Allenoates substituted on the proximal or terminal position were also explored. However, the expected products both couldn't be formed. An isomerized product **5b** from **2b** was produced in 30% NMR yield in the reaction of **1a** with **2b** (eq.15). Meanwhile, the reaction of **1a** with **2c** only afforded Benzoin product **6a** in 3% yield together with some recovery of the reactants (eq.16).



Conclusions

In conclusion, mechanistic studies show that *N*-heterocyclic carbene may react with the aldehydes in the presence of water to afford the monohydrated Breslow intermediate and the solvent $CHCl_3$ acts as the proton shuttle. Subsequent thermodynamically controlled Michael addition with butadienates, repeated protonation and deprotonation involving $CHCl_3$ and water, and elimination yielded the final products. Based on this, we have also developed an efficient reaction for the preparation of (*E*)-4-oxo-2-butenoates from readily available aromatic aldehydes and butadienoates with a high stereoselectivity and atom economy. Further investigation including reactions of other electron-deficient allenes catalyzed by NHC is being carried out in our laboratory.

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Conflicts of interest

There are no conflicts to declare.

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21. Crystal data for compound (*E*)-**3ha**: $C_{20}H_{18}O_4$, MW = 322.34, monoclinic, space group C 1 2/c 1, final R indices [I > 2σ(I)], R1 = 0.0554, wR2 = 0.1329; Rindices (all data), R1 = 0.0817, wR2 = 0.1558; a = 22.7223 (19) Å, b = 5.7888 (4) Å, c = 25.159 (2) Å, α= 90.00°, β= 95.416 (8)°, γ= 90.00°, V = 3294.6 (5) Å^3, T = 293 (2) K, Z = 8, reflections collected/unique 6328/2071 (Rint = 0.0383), number of observations [> 2σ(*I*)] 2997, parameters: 219. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Centre, CCDC 1540041.

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